

ADC-aware PTQ and Post-ADC Low-Rank Correction for Llama

Analog-Digital Compute · Post-Training Quantization

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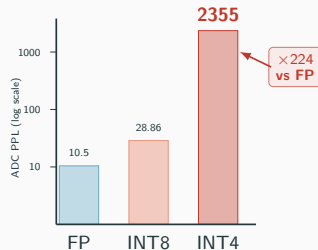
Problem: ADC Severely Degrades Low-Bit LLM Inference

The opportunity

- Analog MVM is 10–100 $\times$ more energy-efficient than digital
- Attractive for LLM inference at scale

The bottleneck: ADC

- Each tile output must be digitised by an ADC with step δ
- Small outputs fall into the **ADC dead band**



Goal

Recover LLM quality under a **frozen analog hardware model** — no changes to ADC circuitry or tile structure.

Research Questions

Q1 — Diagnosis

Is the dead-zone collapse hardware-fundamental, or a transform-training artefact?

Q2 — INT8 recovery

How much can ADC-aware PTQ recover compared to the FP baseline?

Q3 — INT4 pure PTQ

What is the best calibration recipe under INT4 ADC constraints?

Q4 — Breaking the plateau

If PTQ saturates, where and how should residual correction be inserted?

Our approach

1. Improve FlatQuant training
2. Propagated ADC-aware PTQ (α -mixed loss, staged diag)
3. Post-ADC residual LoRA with CE+KL teacher

Q1 → Slide 7 Q2 → Slide 8

Q3 → Slides 9–10 Q4 → Slides 12–14

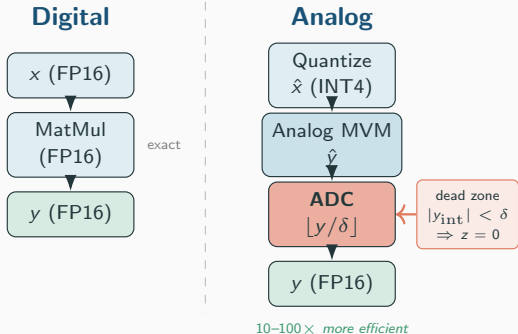
Why Analog Inference? Why This Is Hard.

The opportunity: analog MVM

- LLM inference is memory- and compute-bound
- Analog *multiply-accumulate* is 10–100× more energy-efficient vs. digital
- Real benefit only if output precision is sufficient

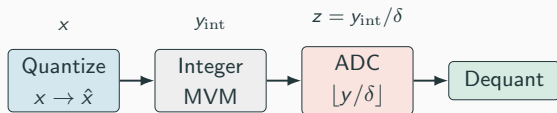
The bottleneck: ADC

- After each tile MVM, an ADC digitises the analog accumulator
- ADC resolution is *coarse*: $\delta \approx 2016$ (INT8 tile)
- Poor activation distributions $\Rightarrow$ most outputs fall in the **dead zone** ($|y_{\text{int}}| < \delta$)



Hardware Model & Metrics

One tile of analog MVM



Dead zone: $|y_{\text{int}}| < \delta \Rightarrow z = 0$ (ADC outputs nothing)

$$\delta = \frac{2 \text{tile_in} \cdot q_x q_w}{2^{b_a} \cdot k} \quad \text{INT8 } (q=127): \approx 2016; \text{ INT4 } (q=7): \approx 6.12$$

Key metrics

- $\text{PPL}_{\text{bypass}}$ — PPL with ADC disabled (transform quality proxy)
- PPL_{ADC} — PPL with full ADC pipeline (what matters)
- `dead_rate` — fraction of channels with $z = 0$

Parameter	Value
Model	Llama-3.2-1B
b_w, b_x	4 or 8 bits
b_a (ADC)	8 bits
Tile size	256
k (parallelism)	16
δ (INT4)	≈ 6.12
δ (INT8)	≈ 2016

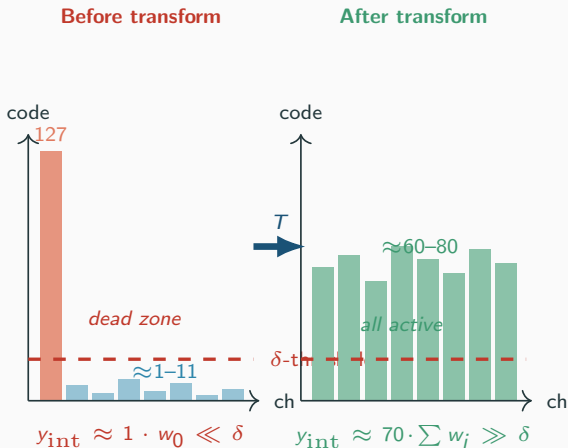
Why FlatQuant Transforms Matter for ADC

Without transform: sparse codes

- Per-token scale $s = \max(|x_{\text{tile}}|)/127$
- One large feature gets code ≈ 127 , rest get codes $\approx 1-10$
- Dot product: $y_{\text{int}} \approx 1 \cdot w_{\text{max}} \approx \text{small} \Rightarrow$ **dead zone**

With FlatQuant transform

- Kronecker-decomposed orthogonal T learned per layer
- After rotation: all features have similar magnitude
- Codes $\approx 50-100$ for all channels
- $y_{\text{int}} \approx 64 \cdot \sum w_i \gg \delta \Rightarrow$ no dead zone



Key proxy: $\text{PPL}_{\text{bypass}}$ = transform quality. Lower $\text{PPL}_{\text{bypass}} \Rightarrow$ uniform codes $\Rightarrow$ less dead zone.

First Key Discovery: Dead Zone Was Not Hardware-Fundamental

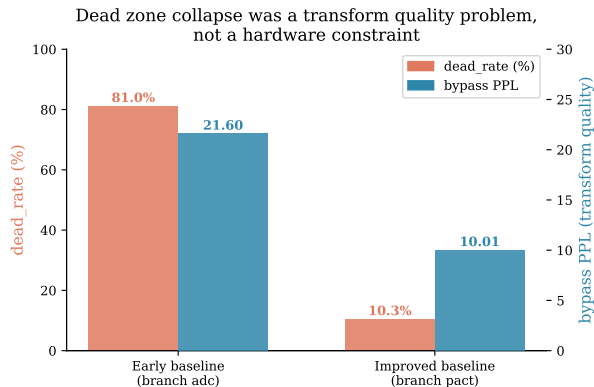
Early baseline (branch

llama-flatquant-adc)

- $\text{PPL}_{\text{bypass}} = 21.6$, $\text{PPL}_{\text{ADC}} = 28.99$
- $\text{dead_rate} = 81\%$
- PACT, penalties, L1 loss $\Rightarrow$ all failed

Improved baseline (branch pact)

- $\text{PPL}_{\text{bypass}} = 10.0$, $\text{PPL}_{\text{ADC}} = 28.86$
- $\text{dead_rate} = 10.3\%$
- **Hardware unchanged.** Only better FlatQuant training (percentile calibration vs. absmax)



Implication: Dead-zone was a **transform quality problem**, not a hardware constraint. Remaining INT8 gap ($\text{PPL}_{\text{bypass}}=10 \rightarrow \text{PPL}_{\text{ADC}}=29$) is **ADC resolution** — only $\sim 15\text{--}30$ useful bins in $\pm\sigma_z$.

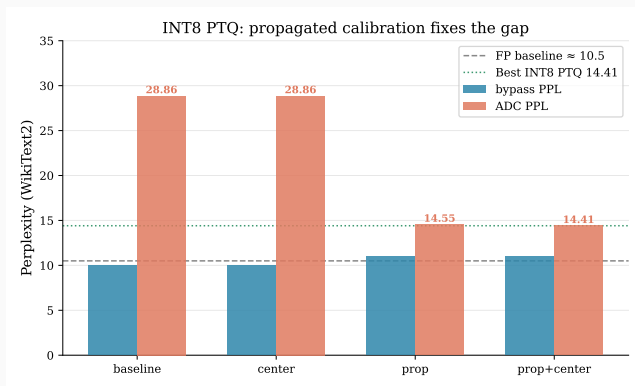
INT8: Propagated Calibration Fixes the Real Problem

Two ideas for closing the $10 \rightarrow 29$ gap:

1. **Bin-centre loss** $\cos^2(\pi z)$:
nudge y_{int} toward bin centres.
Result: no improvement.
2. **Propagated calibration**:
train layer i on *ADC-quantised* inputs
from layer $i - 1$
(not clean FP inputs).
Result: -47% ADC PPL.

Why propagation works

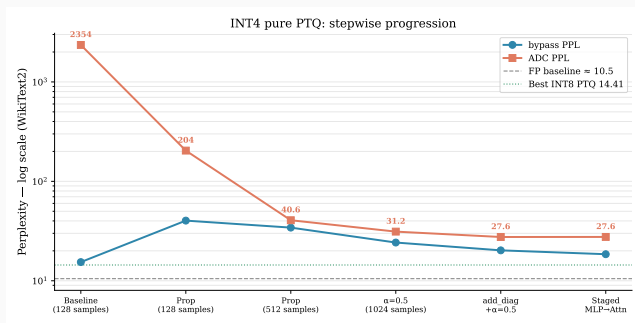
Standard layer-wise PTQ with FP inputs *underestimates* the reconstruction difficulty at inference, where upstream ADC errors accumulate across layers.



INT4 Pure PTQ: A Stepwise Story

Each step fixes one failure mode:

1. **Baseline INT4 (128 samples):**
 $\text{PPL}_{\text{ADC}} = 2355$ — catastrophic; 128 samples insufficient.
2. **Propagation + 1024 samples:**
 $\text{PPL}_{\text{ADC}} = 40.6$ — scale and propagation are the dominant levers.
3. **Partial propagation $\alpha = 0.5$:**
 $\text{PPL}_{\text{ADC}} = 31.2$ then **diag scaling**
→ 27.56 — mixed FP+quant loss + bounded diagonal.
4. **Staged training (MLP → attn diag):**
 $\text{PPL}_{\text{ADC}} = 27.60$, $\text{PPL}_{\text{bypass}} = 18.50$ — cleanest bypass yet.



Key insight

Bin-centre loss is **harmful** for INT4 — fine ADC step ($\delta \approx 6.12$) makes the penalty dominate and destabilise training.

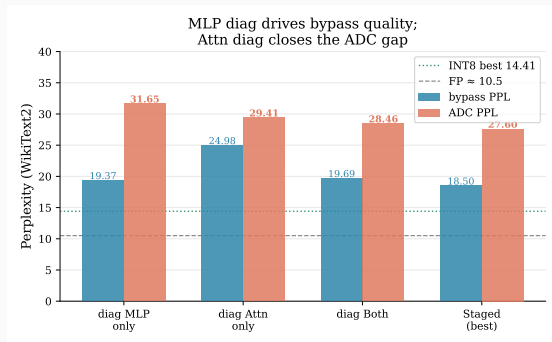
Mechanistic Insight: MLP and Attention Play Different Roles

Finding (v3/v4 ablation):

- **MLP diag** → best *bypass* PPL, but wide ADC gap
- **Attn diag** → closes the ADC gap, doesn't improve bypass
- **Staged training:** Stage A (no prop) learns clean MLP transforms; Stage B (10ep, $\alpha=0.5$) adds attn diag for ADC robustness

Result

Staged MLP→attn: bypass=**18.50**, ADC=**27.60**
Best purely PTQ result — cleanest bypass yet.

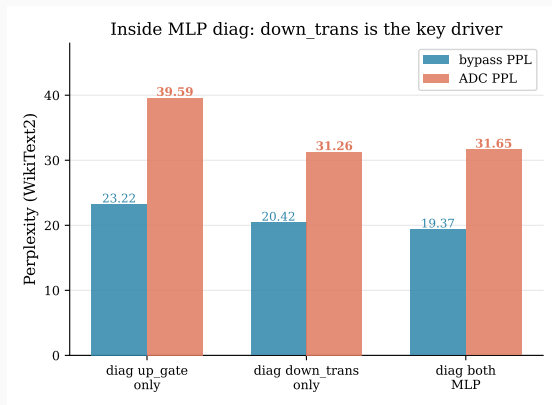


Inside MLP: down_proj Drives ADC Quality

MLP diag split (v4):

- down_trans **diag**: bypass=20.42, ADC=31.26
≈ full MLP diag — this is the key driver
- up_gate_trans **diag**: bypass=23.22, ADC=39.59
much weaker effect
- down_proj sits right before the ADC accumulation
— its scale directly controls $z = y_{\text{int}}/\delta$

Takeaway: Apply diagonal scaling to down_proj first;
up_gate is secondary.



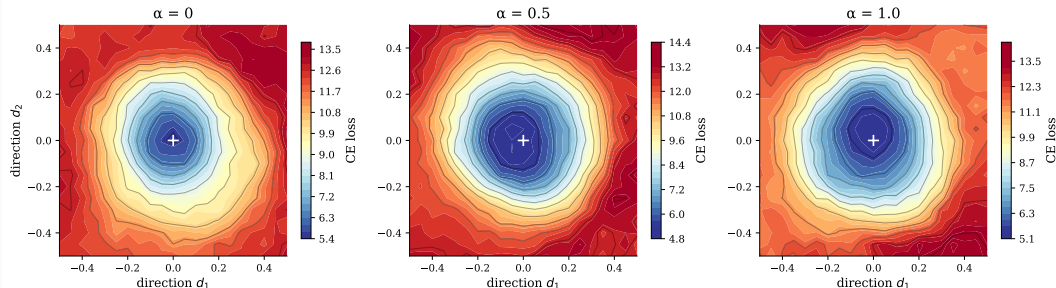
Loss Landscape: Why $\alpha=0.5$ Is the Sweet Spot

Dual-forward mixing loss:

$$\mathcal{L} = (1-\alpha)\mathcal{L}_{\text{FP}} + \alpha\mathcal{L}_{\text{ADC}}$$

- $\alpha=0$: clean FP loss only $\Rightarrow$ flat bypass, wide ADC gap
- $\alpha=1$: transforms over-specialise, bypass degrades sharply
- $\alpha=0.5$: smooth, wide minimum in both metrics

Loss landscape: FlatQuant INT4, varying propagation α



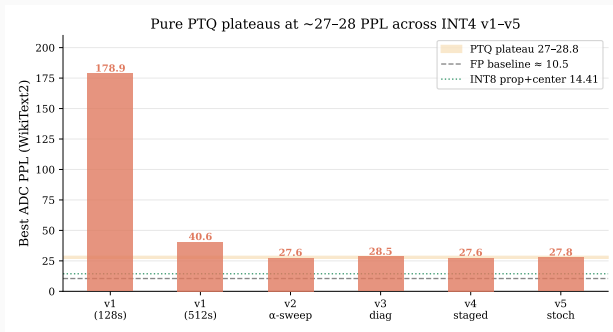
Pure PTQ Hits a Plateau — a Different Tool Is Needed

After all PTQ improvements (v2 through v5):

- Best $\text{PPL}_{\text{ADC}} \approx 27.5\text{--}28.5$ across all INT4 variants
- Adding more calibration samples / stochastic propagation / Hadamard gives < 0.5 PPL gain
- **Root cause:** 4-bit weight quantisation introduces systematic residual error that transform calibration cannot recover — transforms redistribute information, but cannot add it back after INT4 rounding

Key realisation

We need a **post-hoc correction** that can learn to model the ADC residual *after* quantisation, without re-training the frozen PTQ checkpoint.

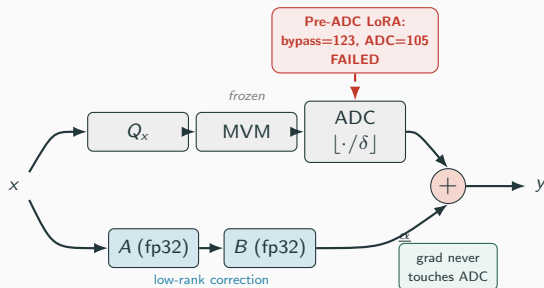


Post-ADC Residual LoRA: Breaking the Plateau

Architecture:

$$y = \underbrace{\text{ADC}_{\text{frozen}}(x)}_{\text{PTQ path}} + \frac{\alpha}{r} B(A(x_{\mathbb{R}}))$$

- Correction applied **after** ADC output
- Base model frozen in fp16; LoRA adapters in fp32
- Gradients never touch `round_ste` / ADC clamp $\Rightarrow$ stable
- Trained with LM cross-entropy (+ optional KL teacher), 5 epochs



Pre-ADC LoRA failed

`pre_adc_r4_down.o`: `bypass=123, ADC=105`.
Gradients through `rnd_ste( $\mathbf{W} + \Delta\mathbf{W}$ )` are too noisy from step 0.

ADC-LoRA Ablations: CE+KL Loss and Rank

1. CE+KL consistently beats CE

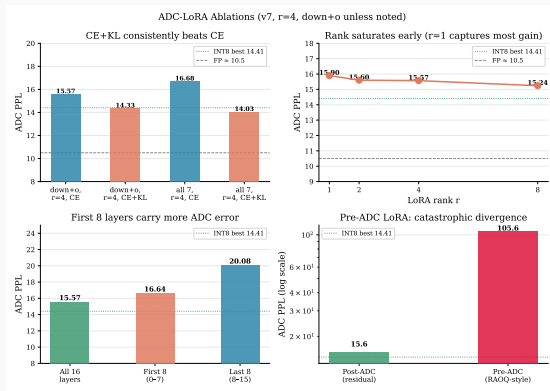
- down+o, r4: CE $\rightarrow$ 15.57; CE+KL $\rightarrow$ **14.33**
- all 7, r4: CE $\rightarrow$ 16.68; CE+KL $\rightarrow$ **14.03**
- FP teacher provides signal CE alone cannot

2. Rank saturation at $r=1$

- $r=1$: 15.90; $r=4$: 15.57; $r=8$: 15.24
- Only 0.66 PPL gain across full rank range
- Most correction capacity captured at rank=1

Best

r4_all_ce_kl: ADC = **13.96** (both domains)



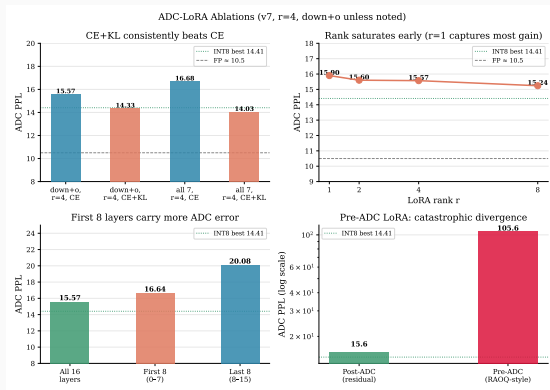
ADC-LoRA Ablations: Layer Selection and Placement

3. Early layers carry more ADC error

- first8: ADC=16.64 last8: ADC=20.08
- Early layers accumulate and propagate errors downstream
- All 16 layers ($r=4$, CE+KL): **13.96** — best result

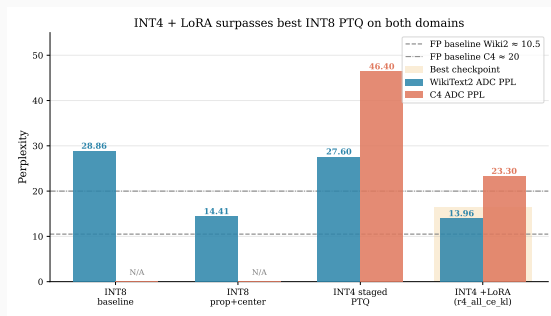
4. Post-ADC is the only stable placement

- Pre-ADC: bypass=123, ADC=105 — **catastrophic**
- Gradients through round_ste too noisy from step 0
- Post-ADC: stable from epoch 1



Final Results: INT4 + LoRA Surpasses INT8 PTQ

Method	Bits	Wiki2	C4
FP baseline	—	~10.5	~20.0
INT8 baseline	w8a8	28.86	—
INT8 prop+center	w8a8	14.41	—
INT4 pure PTQ best	w4a4	27.56	—
INT4 staged PTQ	w4a4	27.60	46.40
INT4 + LoRA	w4a4	13.96	23.30

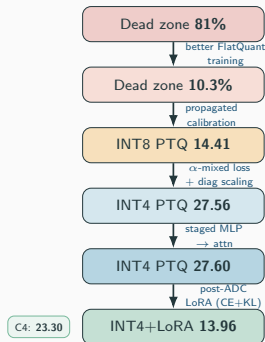


Key messages

- **INT4 + LoRA surpasses best INT8 PTQ on WikiText2** (13.96 vs. 14.41)
- **Generalises to C4** (−50% vs. INT4 PTQ baseline): correction is not overfit to calibration domain

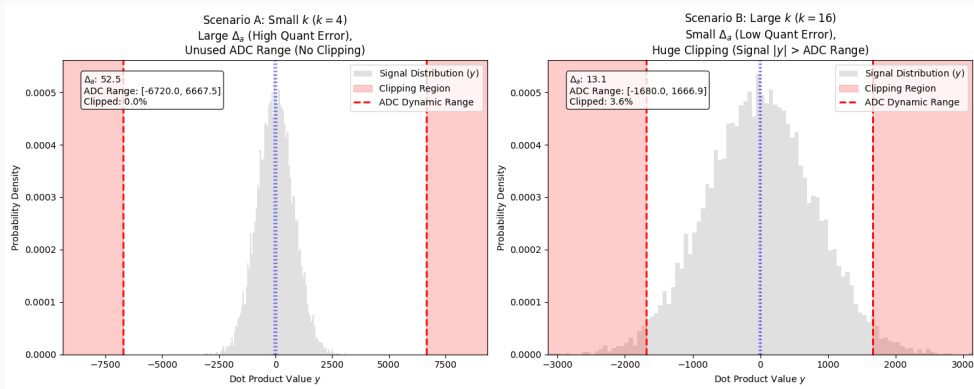
Takeaways

1. **Dead zone was a training problem, not hardware.**
Better FlatQuant calibration: 81% $\rightarrow$ 10.3% dead rate.
2. **Propagated calibration fixes INT8.**
ADC PPL: 28.86 $\rightarrow$ 14.41 (-47%), near FP baseline.
3. **Staged PTQ closes the INT4 gap to $\times 1.4$.**
 α -mixed loss + MLP-first diagonal training.
4. **MLP drives bypass; attn drives ADC robustness.**
Staged training combines both — insight, not just tuning.
5. **Post-ADC LoRA breaks the PTQ plateau.**
INT4 ADC 13.96 < INT8 PTQ 14.41. Generalises to C4 (-50% vs PTQ baseline).



*The right place to correct ADC error is **after** ADC, not before it.*

Backup: ADC Dead Zone Illustration



Backup: Complete INT8 Results (v2)

Experiment	Description	PPL _{bypass}	PPL _{ADC}	dead_rate
baseline	FlatQuant w8a8	10.01	28.86	10.3%
center mid	bin-centre $\lambda = 0.1$	10.01	28.86	10.3%
prop	propagated calibration	11.01	14.55	10.3%
prop+center	propagated + bin-centre	11.00	14.41	10.3%

Backup: Complete INT4 v2 Alpha-Sweep

Experiment	α	PPL _{bypass}	PPL _{ADC}	gap
baseline_1024	0.0	19.80	56.83	×2.87
prop_full_1024	1.0	30.32	40.20	×1.32
propalpha_075_1024	0.75	27.46	35.92	×1.31
propalpha_05_1024	0.5	24.24	31.22	× 1.29
propalpha_025_1024	0.25	21.65	32.35	×1.49
add_diag + $\alpha=0.5$	0.5	20.23	27.56	×1.36
staged_mlp_attn	0.5	18.50	27.60	×1.49

Backup: Complete v7 LoRA Ablation

Experiment	Targets	Loss	Rank	PPL _{bypass}	PPL _{ADC}
base (no LoRA)	—	—	—	17.89	27.55
r1_down_o	down+o	CE	1	—	15.90
r2_down_o	down+o	CE	2	—	15.60
r4_down_o	down+o	CE	4	18.68	15.57
r8_down_o	down+o	CE	8	—	15.24
r4_down_o_ce_kl	down+o	CE+KL	4	19.54	14.33
r4_down_o_first8	down+o	CE	4	18.42	16.64
r4_down_o_last8	down+o	CE	4	18.62	20.08
pre_adc_r4_down_o	down+o	CE	4	123.34	105.63
r4_all_ce	all 7	CE	4	17.62	16.68
r4_all_ce_kl	all 7	CE+KL	4	19.13	14.03

Backup: v8 Generalisation to C4

Experiment	Targets	Wiki bypass	Wiki ADC	C4 bypass	C4 ADC
base (no LoRA)	—	18.49	27.26	29.41	46.40
rank8_down_o (CE)	down+o	18.51	15.47	30.48	29.39
r4_down_o_ce_kl	down+o	19.05	14.27	29.37	23.88
r4_all_ce_kl	all 7	18.70	13.96	29.10	23.30

Backup: Stochastic Propagation (v5)

Experiment	Mode	$\text{PPL}_{\text{bypass}}$	PPL_{ADC}	gap
stoch_bern (flat)	Bernoulli	19.25	35.38	$\times 1.84$
stoch_beta2 (flat)	Beta(2,2)	22.90	31.40	$\times 1.37$
staged + Bernoulli	Bernoulli	16.94	32.44	$\times 1.92$
staged + Beta(2,2)	Beta(2,2)	17.16	27.82	$\times 1.62$
deterministic $\alpha=0.5$	—	24.24	31.22	$\times 1.29$

Note

Bernoulli gives best bypass but worst ADC gap (hard 0/1 switching provides no ADC signal on FP batches). Beta(2,2) $\approx$ deterministic $\alpha=0.5$ in expectation.

Backup: Hardware Config & Delta Formula

Parameter	Value
b_w (weight bits)	4 / 8
b_x (act. bits)	4 / 8
b_a (ADC bits)	8
mvm_limit	256 / 1024
k (ADC parallelism)	16
δ (mvm=256, INT8)	≈ 2016
δ (mvm=256, INT4)	≈ 6.12
δ (mvm=1024, INT8)	≈ 8064
FlatQuant epochs	30
FlatQuant LR	0.005
Cal. samples (INT4)	1024
Cal. samples (INT8)	128

Delta formula ($q = 2^{b-1} - 1$):

$$\delta = \frac{2 \cdot \text{tile_in} \cdot q_x \cdot q_w}{2^{b_a} \cdot k}$$

- INT8 ($q=127$, mvm=256): $\frac{2 \cdot 256 \cdot 127^2}{256 \cdot 16} \approx 2016$
- INT4 ($q=7$, mvm=256): $\frac{2 \cdot 256 \cdot 7^2}{256 \cdot 16} \approx 6.12$
- INT8, mvm=1024: $\frac{2 \cdot 1024 \cdot 127^2}{256 \cdot 16} \approx 8064$

INT8: useful ADC bins in $\pm\sigma_z \approx 15\text{--}30$ (empirical) — explains why INT8 PTQ alone plateaus.

This Work in Context

Method	Key idea	Gap / limit
FlatQuant	Learned Kronecker-decomposed orthogonal transforms per layer	FP loss only; no ADC signal
OmniQuant / LET	Learnable per-channel equivalent scaling (diagonal)	No ADC-aware objective
QEP / prop. calib.	Feed quantised activations into the next layer's calibration	INT8 only; no mixed-loss
PACT / QDrop	Stochastic / clamped quant-aware training	Full re-training required
This work	ADC-aware PTQ (α -mixed, staged diag) + post-ADC LoRA	—

Key novelty

No prior method directly targets the **post-ADC residual** — our LoRA correction is the first to do so.

Post-ADC LoRA: Parameter and Compute Overhead

Trainable parameter count ($r=4$, all 7 projections):

Projection	A	B
q_proj	4×2048	2048×4
k_proj	4×2048	256×4
v_proj	4×2048	256×4
o_proj	4×2048	2048×4
gate_proj	4×2048	8192×4
up_proj	4×2048	8192×4
down_proj	4×8192	2048×4

Inference overhead (per forward pass):

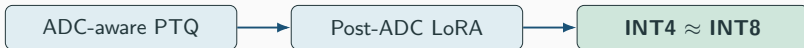
- Extra FLOPs per projection:
 $d_{\text{in}} \cdot r + r \cdot d_{\text{out}} \ll d_{\text{in}} \cdot d_{\text{out}}$
- Example: q_proj: $8192+8192$ vs. $4M$ ($< 0.5\%$ overhead)
- Memory: 2.8M fp32 params $\approx$ **11 MB** vs. ~ 2 GB base model
- LoRA correction runs on *digital* hardware — does not increase analog area

Bottom line

0.28% extra params, $< 0.5\%$ extra FLOPs, correction entirely in digital post-processing. The trade-off is **strongly favourable**.

Thank you!

Questions?



The right place to correct ADC error is after ADC, not before it.

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